FITNESS & HABIT TRACKER

System Architecture & Official Point Allocation Report

1. App Architecture & Workflow

This application is a competitive Habit & Fitness Tracker designed for daily tracking. It uses a modern vanilla HTML/CSS/JS frontend paired with a Google Apps Script backend connected to a centralized Google Spreadsheet database.

- **Frontend (Client-side):** Handles UI interactions, dynamic row rendering, live point calculations, and BMR math natively in the browser to ensure high performance.
- **Backend (Server-side):** Receives JSON payloads, writes them to the respective user's spreadsheet tab, and returns aggregated standing data.
- **Database (Google Sheets):** Contains three main tabs: `Deepak`, `Shalini`, and `CalorieDB`. The CalorieDB acts as a lookup table for exact food and exercise energy values.

2. Calorie Deficit Dashboard

The daily calorie deficit compares Daily Energy Expenditure vs Energy Intake to visually determine if a user is in a healthy deficit (fat loss) or surplus (weight gain).

- **BMR Calculation:** Uses the scientifically verified Mifflin-St Jeor equation.
- **Maintenance Calories:** Base BMR is mathematically multiplied by an **Activity Factor of 1.4** (Sedentary/Active Office Worker) to determine daily maintenance.
- **Deficit Formula:** `Deficit = (Maintenance + Exercise Burned) - Food Consumed`

3. Official Point Allocation Rules

The system features an advanced proportional scoring algorithm. As users fill out their daily entry forms, the system dynamically calculates their score based on the following strict ruleset:

😴 Sleep Tracking

- 6.5 to 7.5 hrs: **+3 pts** (Optimal)
- 5.5 to 6.4 hrs OR 7.6 to 8.5 hrs: **+2 pts**
- 5.0 to 5.4 hrs: **+1 pt**
- Over 8.5 hrs: **-1 pt** (Oversleep Penalty)
- Under 5.0 hrs: **-1 pt** (Poor Sleep Penalty)

📚 Productivity / Study

- 4.0+ hours: **+5 pts**
- 3.0+ hours: **+4 pts**
- 2.0+ hours: **+2.5 pts**
- Under 2 hours: Linear scaling (e.g., 1.5 hrs = **+1.5 pts**)

💪 Exercise (Duration)

- 45+ mins: **+3 pts**
- 30+ mins: **+2 pts**
- 20+ mins: **+1 pt**
- 1 to 19 mins: **+0.5 pts**

📱 Digital Discipline (Phone)

- 0 to 2 hours: **+3 pts** (Excellent)
- 2 to 4 hours: **+1 pt**
- Over 4 hours: **-2 pts** (Penalty)

🥗 Nutrition & Diet

- **Fast/Junk Food:** None (**+2**), Low (**+1**), High (**-1**)
- **Added Sugar:** None (**+2**), Limited (**+1**)
- **Fresh Fruits/Juice:** Consumed (**+1 pt**)
- **Water:** 4L+ (**+3**), 3L+ (**+2**), 2L+ (**+1**), 1L+ (**+0.5**)

🧘 Mental Health & Behavior

- **Meditation:** 20m+ (**+2**), 10m+ (**+1**), 5m+ (**+0.5**)
- **Mood:** Rated 4 or 5 (**+1 pt**)
- **Behavior:** No Anger (**+2 pts**), Showed Anger (**-2 pts**)
- **Sunlight:** 15+ mins (**+0.5 pts**)